H2

MAXIMUS PUNG JUN
H2 Mathematics (9758)
Vectors I
Chapter Test

MATHEMATICS **9758**

Vectors I Test **2026**

1 hour 45 minutes

You must answer on writing paper.

You will need: Writing paper
List of formulae and results (MF27)

INSTRUCTIONS

- Answer **all** questions.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- You should use an approved graphing calculator.
- Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
- Where unsupported answers from a graphing calculator are **not** allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
- You must show all working clearly.

INFORMATION

- The total mark for this paper is 57.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **4** pages.

[Turn over

1 For each of the following statements, state whether it is **true** or **false**.

Each correct answer scores 1 mark. **A blank answer scores 0 marks.** **Each wrong answer scores -1 mark.**

[15]

	Statement	**T/F**
(a)	$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2$.	
(b)	$\lvert\mathbf{a}+\mathbf{b}\rvert^2=\lvert\mathbf{a}\rvert^2+2\,\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2$.	
(c)	$\lvert\mathbf{a}+\mathbf{b}\rvert^2=\lvert\mathbf{a}\rvert^2+2\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert+\lvert\mathbf{b}\rvert^2$.	
(d)	$\mathbf{a}\times\mathbf{a}=\mathbf{a}^2$.	
(e)	$\mathbf{a}\times\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where θ is the angle between **a** and **b**.	
(f)	$\mathbf{b}\times\mathbf{a}=-\mathbf{a}\times\mathbf{b}$.	
(g)	$(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b})=2(\mathbf{b}\times\mathbf{a})$.	
(h)	$(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2$.	
(i)	$\mathbf{a}\cdot(\mathbf{a}\times\mathbf{b})=0$.	
(j)	$\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$.	
(k)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$ and $\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$.	
(l)	For non-zero **a** and **b**, if $\mathbf{a}\cdot\mathbf{b}=0$, then **a** is perpendicular to **b**.	
(m)	If $\mathbf{a}=3\mathbf{b}$, then $\mathbf{a}\times\mathbf{b}=\mathbf{0}$.	
(n)	For non-zero **a**, $\lvert\mathbf{a}\rvert\,\mathbf{a}$ is a unit vector.	
(o)	$\mathbf{a}\times\mathbf{b}$ and $\mathbf{b}\times\mathbf{a}$ point in the same direction.	

2 [DHS 2023 Promo Q5]

With reference to the origin O, the points A and B have coordinates $(1, 2, 0)$ and $(-5, 1, 0)$.

(a) Find the angle AOB. [3]

The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are projected along the direction of $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ to form the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$ respectively.

(b) Find the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$. [3]

(c) Find the exact area of the triangle $O'A'B'$. [2]

3 [NYJC 2024 Promo Q5]

The position vectors of points A, B and C, relative to the origin O are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}, \quad \mathbf{b} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}, \quad \mathbf{c} = 3\mathbf{i} - \mathbf{j} - \mathbf{k}.$$

(a) Find $\mathbf{a} \times \mathbf{b}$ and deduce the exact area of triangle OAB. [3]

(b) Hence, find the volume of the tetrahedron $OABC$.
[The volume of a tetrahedron is $\frac{1}{3} \times$ base area $\times$ perpendicular height.] [3]

(c) Find the perpendicular distance from B to line OA. [2]

4 [MI 2024 Promo Q4]

Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC : AB = 1 : 3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a} \times \mathbf{b}|$, where k is a constant to be determined. [2]

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a} - \mathbf{b}) \cdot \mathbf{c}$. [3]

5 [TMJC 2024 Promo Q6]

(a) The points A, B and C have position vectors $\begin{pmatrix} 1 \\ 3 \\ -7 \end{pmatrix}$, $\begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -9 \\ 8 \\ -27 \end{pmatrix}$, with respect to the origin O.

(i) Show that the points A, B and C are collinear. [2]

(ii) Find the exact area of the triangle OAB. [3]

(b) The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are defined such that $|\mathbf{u}|=1$, $|\mathbf{v}|=2$ and $|\mathbf{w}|=3$.

(i) Suppose that $\mathbf{u}+\mathbf{v}=\mathbf{w}$. With the aid of a diagram, or otherwise, explain why $\mathbf{u}$ is a unit vector that is parallel to $\mathbf{w}$. [2]

(ii) Given instead that each of the three vectors is perpendicular to the sum of the other two vectors, find the exact value of $|\mathbf{u}+\mathbf{v}+\mathbf{w}|$. [4]

6 [RI 2025 Prelim P1 Q3]

Relative to the origin O, the points A and C have position vectors $\mathbf{a}$ and $2\mathbf{b}$ such that

$$\mathbf{a}=5p\mathbf{i}-2p\mathbf{j}+4p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k},$$

where p is a positive constant. It is given that $|\mathbf{a}|=2|\mathbf{b}|$.

(a) Find the exact value of p. [2]

(b) Evaluate $(\mathbf{a}+2\mathbf{b})\cdot(\mathbf{a}-2\mathbf{b})$. [1]

(c) Evaluate $\mathbf{a}\times\mathbf{b}$ and hence find the area of triangle OAC. [3]

(d) Use a geometrical reason to explain why $|\mathbf{a}\times\mathbf{b}|=\frac{1}{4}|\mathbf{a}+2\mathbf{b}||\mathbf{a}-2\mathbf{b}|$. [2]

END OF TEST